

DocPipe

Scans in, trusted and searchable credit files out

Governed document intake for corporate credit: a vision model reads, our checks decide, a person approves what is uncertain.

TuanBC · Techcombank AI Transformation · Expert Challenge 2026

Recommendation: a 6-week shadow pilot with one credit team

A vision model reads every document; deterministic checks, a review queue and an audit trail decide what is trusted. The model is the cheap, swappable part; the controls are ours.

16/16

synthetic scans: type, customer and year all correct (incl. 6 traps)

106/106

printed amounts and codes read exactly

0

wrong results auto-filed

~870 h

per month saved at base volumes (ESTIMATE, 5.5 FTE)

The ask

- Weeks 1-2: interviews, timed baseline, in-bank model endpoint decision, legal review
- Weeks 3-6 shadow mode, weeks 7-12 A/B by batch; stop rules signed by risk before day 1
- Measured on real scans before any scale-up: synthetic results prove the flow, not the accuracy

A credit file is a pile of scans that people read, file and re-key

Time

A financial spread takes 30 min to 6 h per borrower (public vendor range). Each onboarding document is minutes, and there are many per customer.

Trust

A keyed number has no link back to its page. A dropped digit can reach a credit memo, and nobody can replay where it came from.

Re-finding

"What is the collateral on this facility?" means asking a colleague or reopening folders with inconsistent names.

Documents in scope: financial statements and annual reports, business registration, board resolutions, contracts, invoices, collateral papers, tax filings, bank statements.

Problem statement: credit staff spend hours reading, filing and re-keying documents a machine can read, and the result is neither traceable nor searchable.

Four roles, three conflicts the design must settle



Analysts, RMs

Wants: Less keying; numbers they can trust **Fears:** Checking costs as much as doing



Credit operations

Wants: Throughput, standard folders, SLAs **Fears:** A queue that must be staffed



Risk, compliance

Wants: Page-level traceability; no data leaving **Fears:** Silent errors; unreviewed model changes



Technology

Wants: Swappable parts, clear licences **Fears:** Lock-in; an AGPL surprise

Conflicts and how the design responds

Speed vs traceability

Every value carries its page; checks gate auto-filing

Capability vs data control

Hosted model for public data only; in-bank endpoint is a gate

Automation vs verification cost

Reviewers see only flagged documents; review rate is alerted on

Evidence status: no interview yet; needs are from cited desk research: 70% of financial institutions lost clients to slow onboarding (Fenergo 2025); spreads take 2-6 h (vendor range); PDPL fines up to 5% of revenue. Five interviews are the first pilot gate.

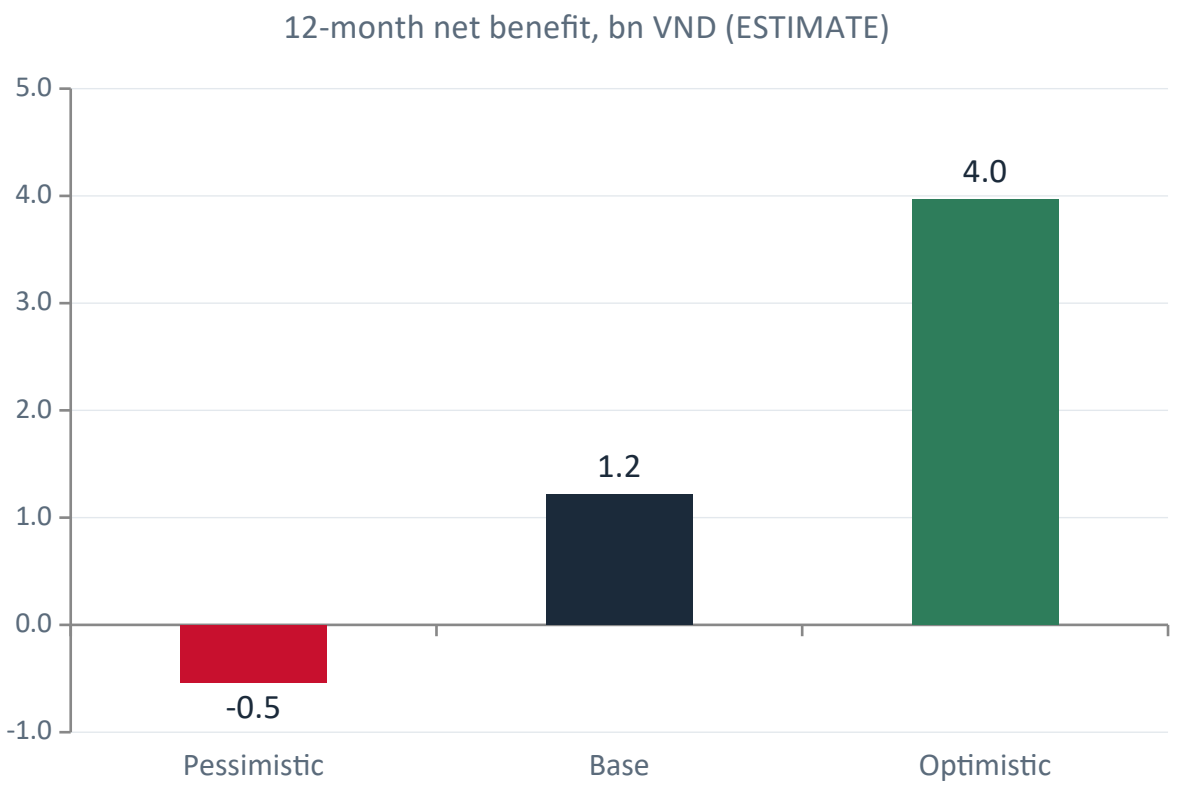
Five directions; the hybrid keeps trust in the bank's hands

Option	Speed	Control and audit	Cost	Lock-in	Verdict
Do nothing	None	Manual, inconsistent	Grows with volume	None	Baseline
Staff use a chat assistant	Fast	None; data leaves	Low	Medium	Rejected: shadow IT
Buy a document-AI suite	Fast start	Vendor-defined; VN statements unproven	Licence + per page	High	Bake-off on our eval set
Build own OCR + layout models	Slow	Full	High: GPUs, templates	Low	Over-engineered
Hybrid: model reads, our checks decide	**Fast**	**High: rules, review, audit ours**	**Model ≈ 0; people dominate**	**Low: swappable**	**Chosen**



Reading text is now the cheap, replaceable part. The scarce part is trust: checks, evidence, a review path and an audit trail. Keep those; swap the model (hosted today, in-bank tomorrow).

~1.2 bn VND net in 12 months, if the baseline holds



Scenario inputs: pessimistic 30% auto-filed, 30/3 manual min · base 60-70%, 60/6 min · optimistic 85%, 120/10 min. Model cost ≈ 0.2 m VND a month.

Base case (ESTIMATE)

- 600 statement files + 4,000 documents a month (assumed)
- ~870 hours saved, 5.5 FTE of capacity
- Payback ~5.1 months after a 0.9 bn VND one-off

If the assumptions are wrong

Manual minutes move the result more than automation: at 30/3 min the case is negative at every automation level (20-80%); at 60/6 min it is positive at every level. So week 1 times the real work and the case is re-run before anything scales.

A model reads, our checks decide, a person approves

Upload

PDF, scan or phone photo; web UI or API

OCR

Text layer free; image pages to the vision model

Classify

Fixed taxonomy; must quote evidence and name the customer

Check

Evidence in text; statement identities; two reads agree

File

company / year / period-or-type, sidecar JSON, PII masked

Publish + ask

OpenViking tree; answers cite the page

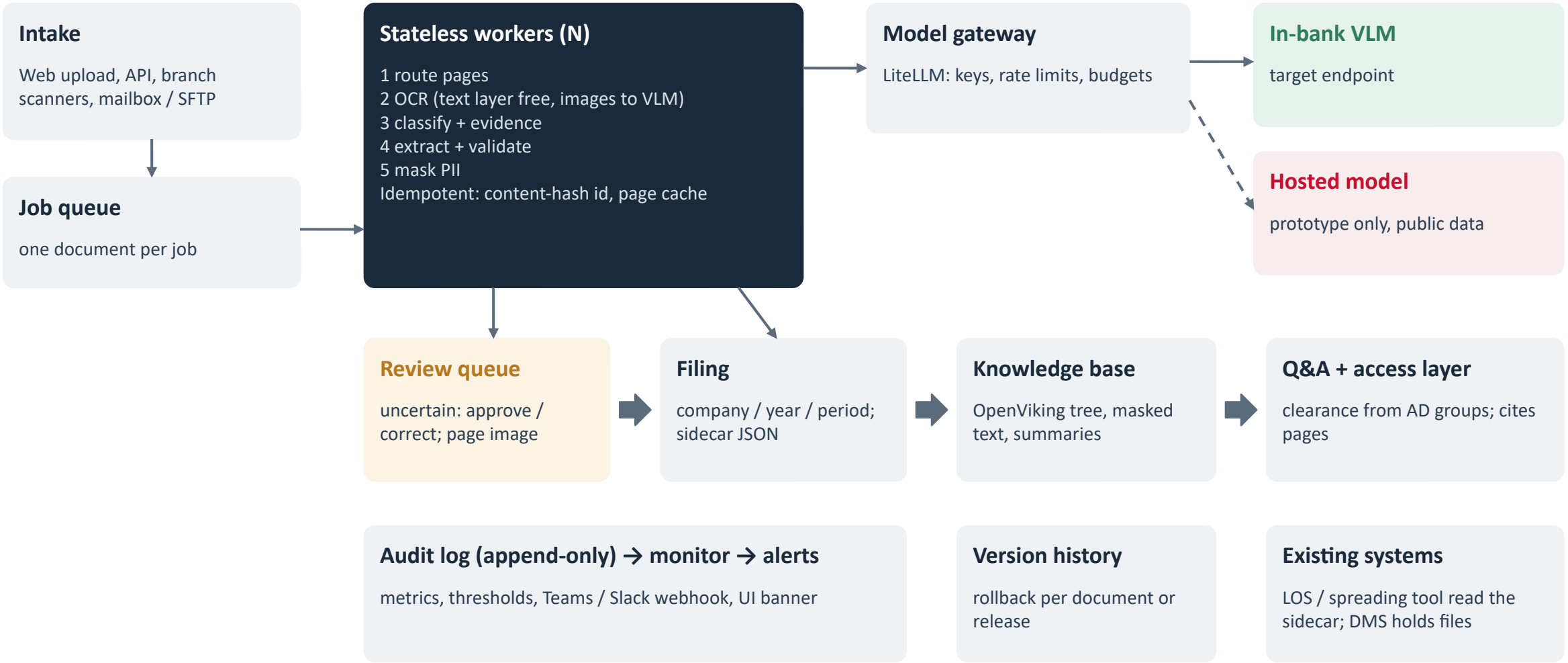
Uncertain? A person decides

Failed check, low confidence, type outside the taxonomy or kill switch on: the document waits in the review queue with its reasons and the page image. Nothing unapproved reaches the knowledge base.

Every step is on the record

Audit log: model, prompt hash, tokens, latency, raw reply, checks, decision, reviewer, correction, rollback, alert. Any filed value can be traced to a page and replayed.

Target architecture: stateless workers, one gateway



Three decisions and what they cost in money, speed and risk

Vision model for OCR, not an OCR engine + rules

Gain: reads diacritics, stamps, skewed photos, tables; no templates. 106/106 numbers exact.

Price: $0.00134 and 13 s per document; can invent text (seen: a table header); images leave the bank if hosted.

Limit: checks, review, in-bank endpoint gate; text layer used free when present.

Checks and people decide, not model confidence

Gain: every auto-filed result passed rules we can explain to an auditor; failures are countable.

Price: correct documents sometimes wait (false alarms on new layouts); reviewer minutes.

Limit: page image next to the value; thresholds from pilot data; review rate alerted.

OpenViking behind an adapter, our own access layer

Gain: viking:// tree mirrors the folders; per-document summaries rank well; agents via MCP.

Price: AGPL-3.0, young project, RBAC unverified; ~40 s summary call per document.

Limit: local BM25 fallback built; citations and access filter stay in our code; legal review.

Security, privacy and audit are designed in, not bolted on

Data classification

Each document tagged internal or restricted from what it publishes. Originals and the PII map stay in a restricted store; the KB gets masked text only.

Access control

Q&A drops chunks above the caller's clearance before the model sees them; a restricted chunk never reached an internal user in tests.

Audit trail

Append-only log of every model call, check, decision, approval, correction, rollback, alert and kill-switch change, with actor and time.

Prompt injection

Document text is data, never instructions. A contract telling the model to reclassify it was filed correctly; Q&A is read-only.

Privacy ladder: a text filter cannot mask an image, so the real barrier is where the model runs

L1 Local regex: Vietnamese IDs, phones, e-mails

Built, measured

L2 + local Vietnamese NER for names, addresses

Not built: names are a known gap

L3 In-bank VLM endpoint: images never leave

Gate before pilot

Measured, not trusted: quality, failures, cost and latency

What	Set	Result
Type, customer, year	16 synthetic scans, 6 traps	16/16 all correct
Printed amounts and codes	same	106/106 exact
Character error rate	same	median 0.013, mean 0.033
Wrong result auto-filed	same	0 (menu and faint receipt went to review)
Injected statement errors	real public filings	6/6 routed to review
Real dropped digit (511.184 → 51,184)	Vinamilk annual report	caught by a rule, corrected, logged
Cited Q&A over filed documents	31 answerable + 6 unanswerable	31/31 cited correctly, 6/6 refused
Cost / latency per document	list price	$0.00134 · 13 s (max 21.8 s)



Honest failure cases

- The OCR model added text: an invented table header and a stamp description (numbers still exact)
- A faint receipt was filed under the payee; a receipt rule is needed
- New issuer layouts caused false alarms until patterns were fixed
- Synthetic scans are easier than branch scans, and I wrote the labels: this proves the flow, not real accuracy

Automated to production standard, human in the loop

Safe to re-run

Content-hash id, per-page cache, re-filing replaces the old copy

Monitored

Review rate, queue age, validator failures, schema errors, p95 latency, cost / doc

Alerts

Thresholds in config; fire once, resolve once; file, audit, UI banner, webhook

Kill switch

One toggle: nothing auto-files, everything to review; logged

Rollback

Every replaced version kept; per document or whole release; reversible

Human review

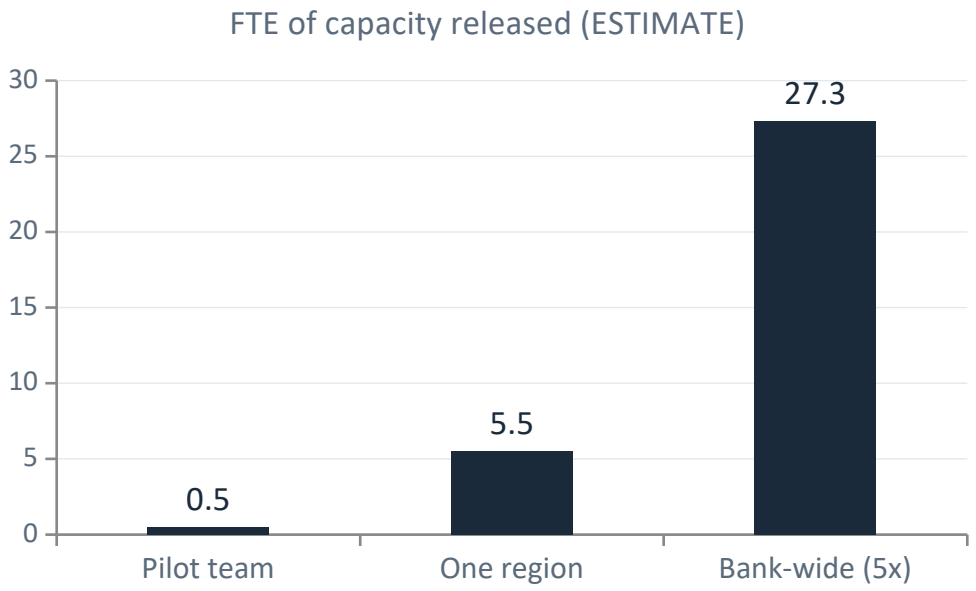
Approve or correct one value; re-validated, re-filed, reviewer logged

Shown on real output

- The monitor fired a real alert: a review ticket had waited 51 h against a 24 h limit
- Rollback drill: a filed invoice restored to v1, re-published to the KB, actor logged, and the quality alarm tripped as designed
- Governance: confidence comes from checks (each failure halves it); threshold 0.90; nothing unapproved is published

Before and after, and projected at scale (estimate)

	Before (manual)	After (DocPipe)
Statement file	60 min (assumed; public 30-360)	30-60 s machine; 15 min review if flagged (~40%)
Onboarding document	6 min (assumed)	13 s machine; 2 min review if flagged (~30%)
Model cost	n/a	$0.00134 per document (measured)
Traceability	Depends on who keyed it	Page, check and log line for every value



Labelled as estimates

- One region (base volumes): ~874 h a month, 5.5 FTE, 1.2 bn VND net in 12 months
- Bank-wide at 5x volume (assumed): ~4,368 h a month, 27.3 FTE; run cost grows with the square root of volume
- A single pilot team does not pay back the one-off (-1.0 bn VND): the case needs at least a region

One North Star, four supporting metrics, shadow then A/B

NORTH STAR

Straight-through correct rate

Share of documents filed with no human touch and no critical error found by the weekly audit sample. Target at day 90: at least 60%.

Handling minutes per document	**–50% vs week-1 baseline**
Critical error rate on auto-filed (5% audit)	**≤ manual and < 0.5%**
Review queue share / oldest ticket	**< 30% / < 24 h**
Questions with a correct page citation	**≥ 90% of 50**

Weeks 1-2

Baseline

Time the manual work; audit its errors

Weeks 3-6

Shadow

DocPipe runs beside the team; nothing used; compare

Weeks 7-12

A/B by batch

Randomised by day: half start from DocPipe, half manual

Later

Stepped wedge

A second team joins: a second comparison

Stop if: audited error rate stays above manual after two tuning cycles · review queue above 50% after week 6 · timed saving under 20% · any PII or access leak · no approved endpoint by week 4

Three months, five people, a gate at every phase

Weeks 1-2

Foundations

Interviews, timed baseline, model endpoint, legal review (AGPL, cross-border)

Gate: Risk signs data handling; case re-run

Weeks 3-6

Shadow pilot

One credit team; weekly audit; 200 real documents labelled; alerts to team channel

Gate: Error ≤ manual; review < 50%

Weeks 7-10

A/B assist

Half the batches start from DocPipe; Q&A on approved documents

Gate: Handling time −20% or better

Weeks 11-13

Harden, scale

Second team by configuration only; rollback drill; audit walkthrough

Gate: Go / no-go for the region

Team: product and process owner · 2 engineers · risk partner (part time) · analyst from the pilot team (part time)

Biggest risks and how we find out early

A wrong value passes every check

Weekly audit sample + 'auto-filed then corrected' alarm; add the missing check

Real scans are harder than the test set

Week-3 eval on 200 real documents; thresholds set on real data

Review burden cancels the savings

Review rate monitored; tune false alarms; stop rule

Built to be reused, and honest about what it is not

Another team adopts it by configuration

- Copy config: taxonomy, model endpoint, thresholds, KB root, alert webhook
- Drop 15-20 labelled samples; run the eval; go live only if wrong-auto-filed is 0
- Same UI, same audit, same rollback: one platform, many teams

What I would not build

- Credit decisioning: a different risk class; DocPipe prepares inputs
- Fine-tuning: no labelled data yet; checks already meet the bar on samples
- A warehouse or a DMS / LOS replacement: DocPipe files into them
- Unsupervised filing outside the taxonomy

Decision requested: approve a 6-week shadow pilot with one credit team, with the in-bank endpoint gate and the stop rules above. I will bring back real-scan accuracy, timed savings and a re-run business case.